FloatLib arithmetic performance from 2 to 4,096 bits

Median end-to-end latency across 9 trials. Lower values are faster.

FloatLib posit — Stillwater Universal posit
FloatLib binary — Berkeley SoftFloat
MPFR (binary) — native C FPU
CPython float64

Addition

Subtraction

Multiplication

Division

Square root

Fused multiply-add

Median latency (ns/operation, log scale)

Encoded width (bits, log₂ scale)

Markers are measured widths. Lines connect those measurements; backend changes are recorded separately in backend-regimes.csv.